NAME : ______________________ CLASS : ______________

JURONG PIONEER JUNIOR COLLEGE
JC2 Preliminary Examination 2022

BIOLOGY
Higher 2

9744/02
13 September 2022

Paper 2 Structured Questions

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
Total	

This document consists of **31** printed pages and **1** blank page.

Answer **all** questions.

1 Fig. 1.1 is an electron micrograph of a cell in the pancreas with structures labelled **X** and **Y**.

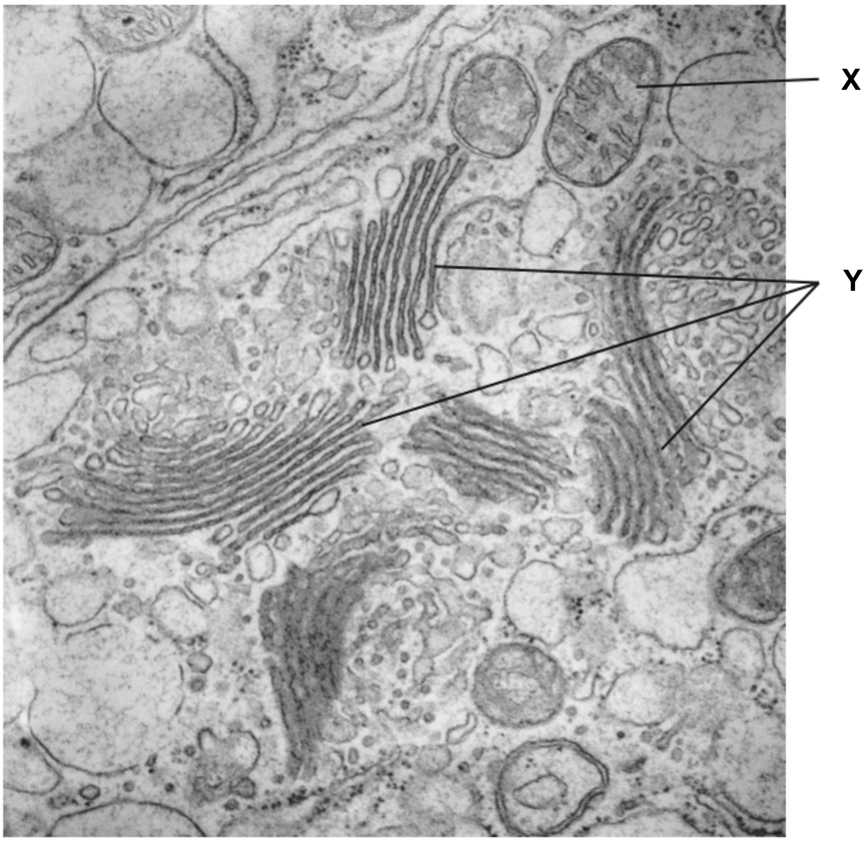

Fig. 1.1

(a) Identify structures **X** and **Y**. In each case, outline one visible feature that allows the structure to perform its functions.

structure **X** ....

visible feature related to its functions ....

....

....

structure **Y** ....

visible feature related to its functions ....

....

....

[4]

Protein production involves a complex sequence of events and number of cell structures. Cells in the pancreas secrete enzymes, such as amylase, into a duct.

(b) The first column in Table 1.1 shows some of the events that occur in the production of amylase in cells in the pancreas and its eventual release from the cell.

Table 1.1

event	sequence of events (numbers)	cell location (letters)
exocytosis		**F**
protein modification		
secretory vesicle formation		
transcription		
translation		

(i) In Table 1.1, write the sequence in which the events occur, using 1 as the first process in the sequence. [1]

(ii) From the list **A** to **F**, choose **one** cell location for each event and write the letter in Table 1.1. Each letter may be used once, more than once, or not at all. The first example, **F**, has been completed for you.

A Golgi apparatus

B lysosome

C nucleus

D rough endoplasmic reticulum

E smooth endoplasmic reticulum

F cell surface membrane [1]

(c) Explain how amylase that are secreted by cells in the pancreas are packaged into vesicles.

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.... [4]

[Total: 10]

2 Fig. 2.1 shows a section of the cell membrane of a red blood cell.

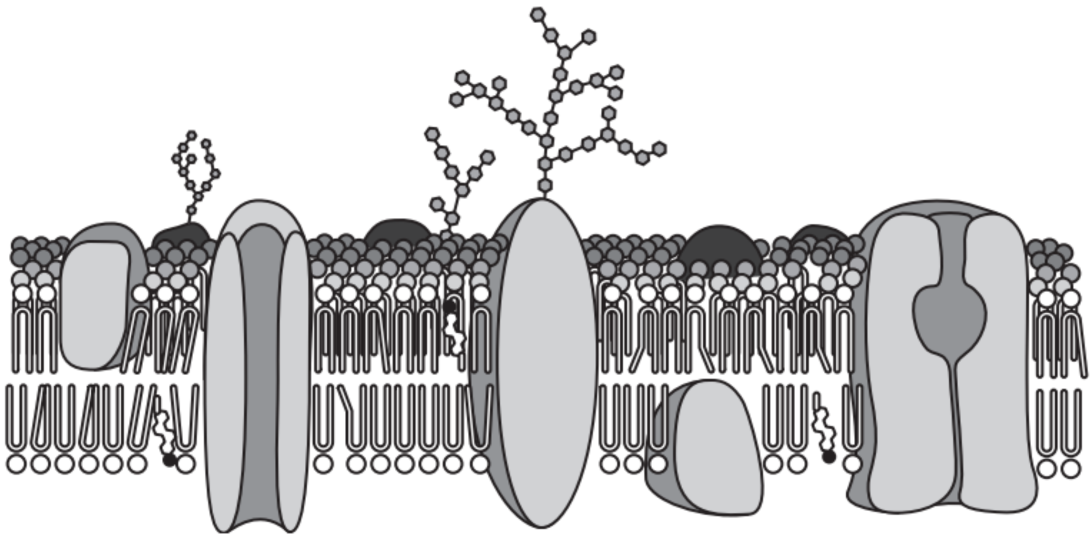

Fig. 2.1

(a) The arrangement of the various constituent biomolecules supports the fluid mosaic model of the cell membrane.

Explain why the cell membrane is referred to as fluid mosaic.

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....

.... [3]

Cyanide is a poison that inhibits cytochrome c oxidase involved in aerobic respiration.

Fig. 2.2 shows how cyanide concentration affects the uptake of chloride ions by red blood cells. The rates of chloride ion uptake are given as percentages of those obtained in a control experiment with no cyanide.

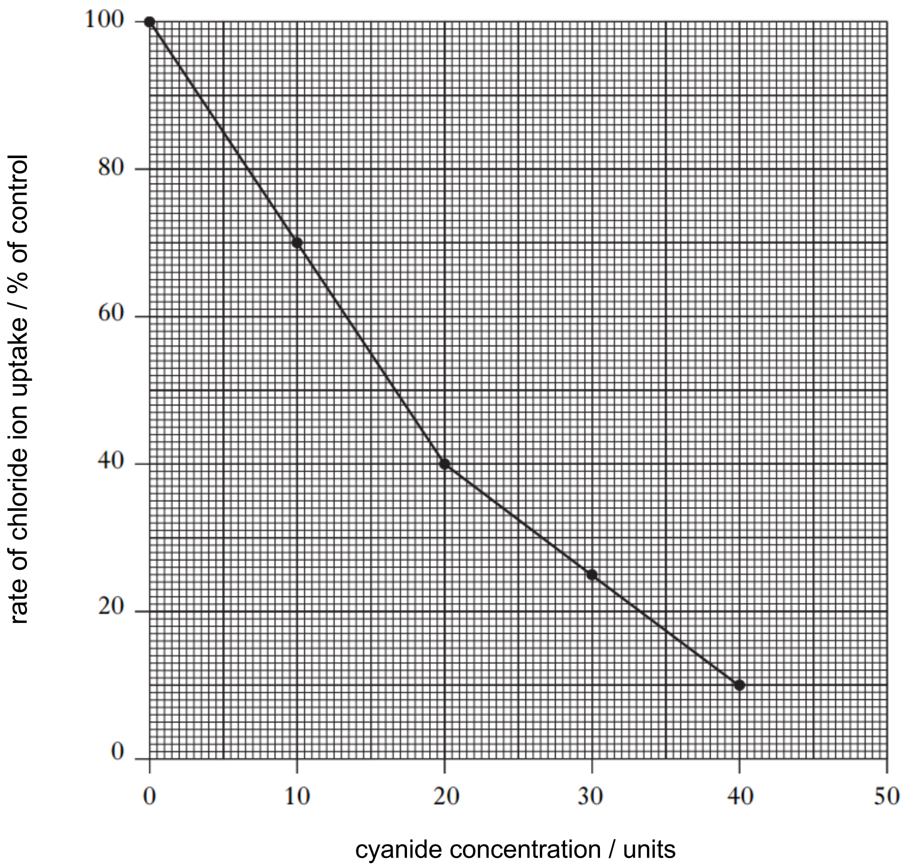

Fig. 2.2

(b) It is known that the uptake of chloride ions requires the presence of transmembrane proteins.

Explain why chloride ions cannot freely pass through the phospholipid bilayer of membranes.

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.... [2]

(c) With reference to Fig. 2.2,

(i) describe how cyanide concentration affects the rate of chloride ion uptake

....

....

....

.... [2]

(ii) identify how chloride ions are moved across the membrane of the red blood cells

.... [1]

(iii) account for your answer in **(c)(ii)**.

....

....

....

.... [2]

[Total: 10]

3 The DNA double helix can be successfully compacted into a nucleus with a diameter that is only a millionth of the length of DNA.

(a) Explain how DNA can be successfully compacted into a nucleus.

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.... [4]

Semi-conservative DNA replication occurs in the nucleus. This begins at the origins of replication, where the enzyme helicase causes the DNA molecule to unwind and unzip, causing the two parental strands to separate.

(b) Outline the role of **three other** enzymes in DNA replication.

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.... [3]

Besides DNA replication, transcription also occurs in the nucleus.

(c) Complete Table 3.1 using a tick (✓) to indicate which features apply to each of the processes. Use a cross (×) for features that do **not** apply. The first row has been completed for you.

Table 3.1

feature	DNA replication	transcription
a single-stranded molecule is produced	×	✓
hydrogen bonds are broken		
both strands of DNA act as templates		
covalent bonds are formed		

[3]

[Total: 10]

4 Fig. 4.1 shows some of the main structural features of an influenza virus. PA, PB1 and PB2 are RNA polymerases.

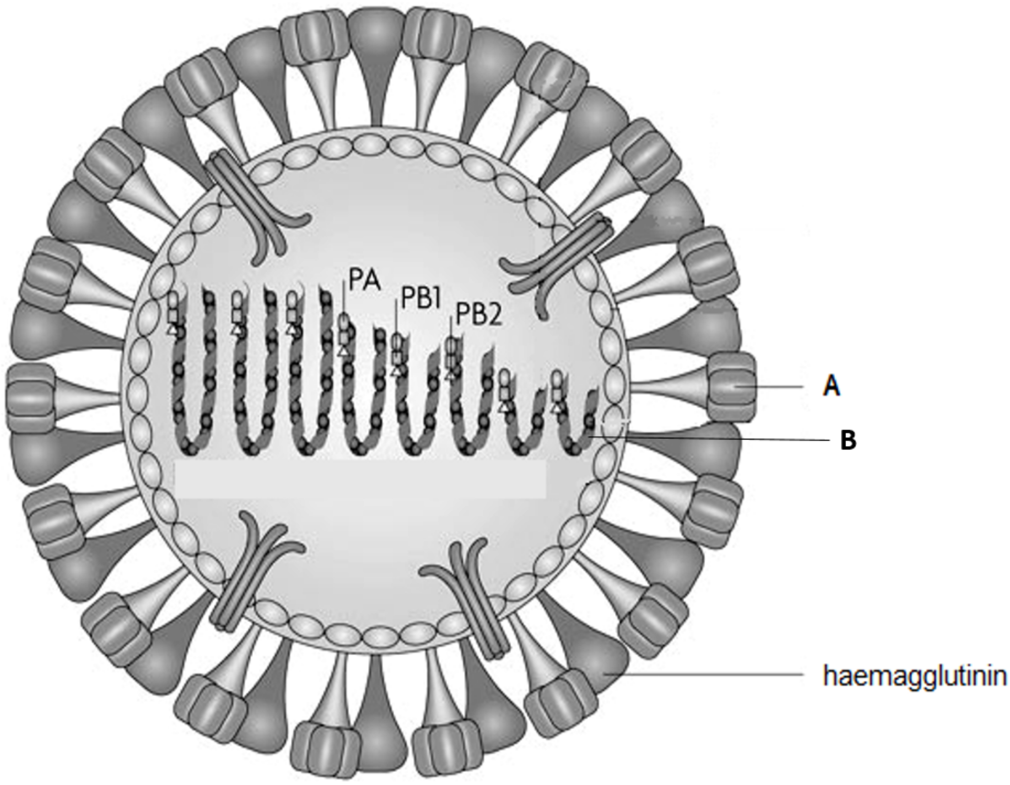

Fig. 4.1

(a) Identify the labelled structures and state their functions.

structure **A** ....

function ....

....

structure **B** ....

function ....

.... [4]

(b) Explain why the influenza virus requires its own RNA polymerases PA, PB1 and PB2.

....

.... [1]

(c) Discuss how viruses challenge the cell theory.

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.... [2]

(d) Oseltamivir is an antiviral inhibitor used for the treatment of infection with influenza viruses.

Fig. 4.2 shows the number of copies of influenza genome per cm^3 plasma in two groups of infected patients – one group treated with oseltamivir and another control group without oseltamivir.

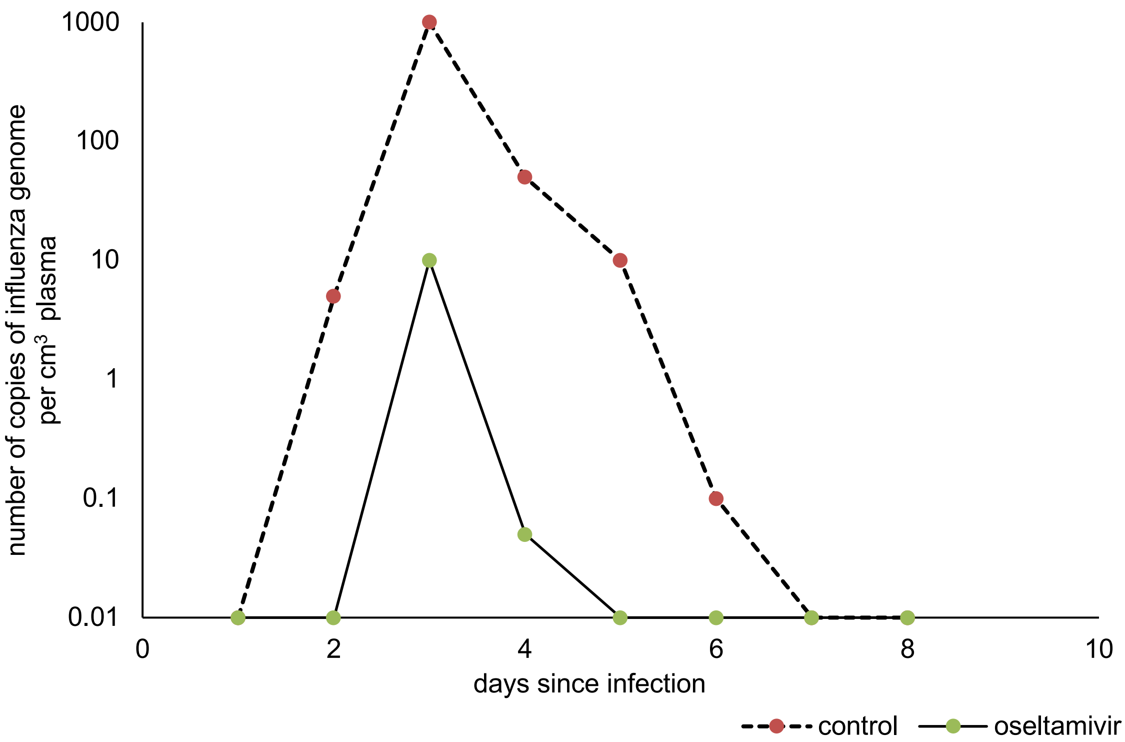

Fig. 4.2

Describe the effect of oseltamivir on the number of copies of influenza genome in patients, as shown in Fig. 4.2.

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.... [3]

[Total: 10]

5 **(a)** Describe **two** ways in which the *lac* operon is **similar** to the *trp* operon.

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.... [2]

(b) Describe **four** ways in which the *lac* operon is **different** from the *trp* operon.

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.... [4]

(c) Suggest the advantages to bacteria of arranging some genes in operons.

....

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.... [2]

[Total: 8]

6 The polymerase chain reaction (PCR) is commonly used in medical and biological research to produce large quantities of DNA from a very small original sample.

(a) The main steps of one PCR method are shown in Fig. 6.1.

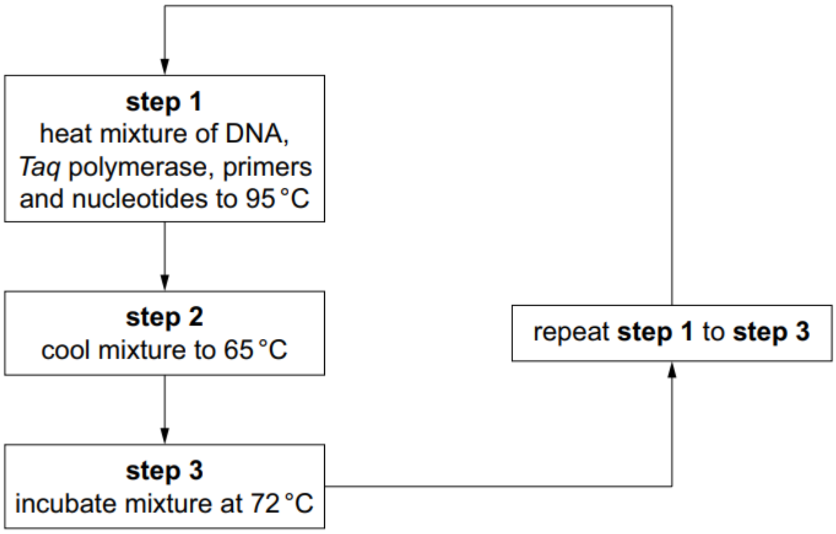

Fig. 6.1

(i) Explain why it is necessary to heat the mixture to 95 °C (**step 1**).

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.... [2]

(ii) Explain why the enzyme *Taq* polymerase, rather than any other type of DNA polymerase, is used in PCR.

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.... [2]

Another molecular technique used in medical and biological research is Southern blotting followed by nucleic acid hybridisation.

To visualise the results, autoradiography is carried out at the end. The images formed on the photographic X-ray film would correspond to the bands that contain the DNA sequence of interest.

(b) Outline the process of Southern blotting and nucleic acid hybridisation.

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.... [4]

Sickle cell anaemia is a genetic disease caused by a recessive mutant allele of the β-globin gene.

Scientists investigated the inheritance of the disease in one family. DNA was collected from each family member. PCR, restriction enzyme digestion using the enzyme *DdeI*, gel electrophoresis, Southern blotting and nucleic acid hybridisation were carried out.

Fig. 6.2 shows the autoradiography results from the final stage of this investigation.

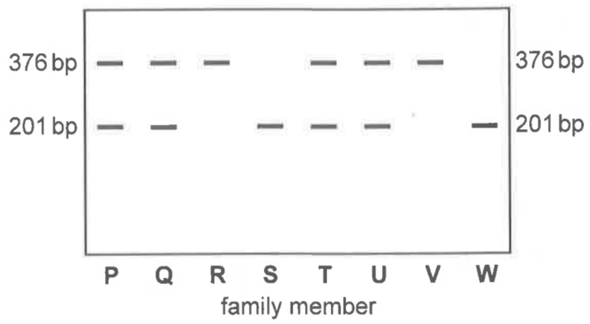

Fig. 6.2

Fig. 6.3 shows where the restriction enzyme *DdeI* cuts within the two different β-globin alleles and the sizes of the fragments produced.

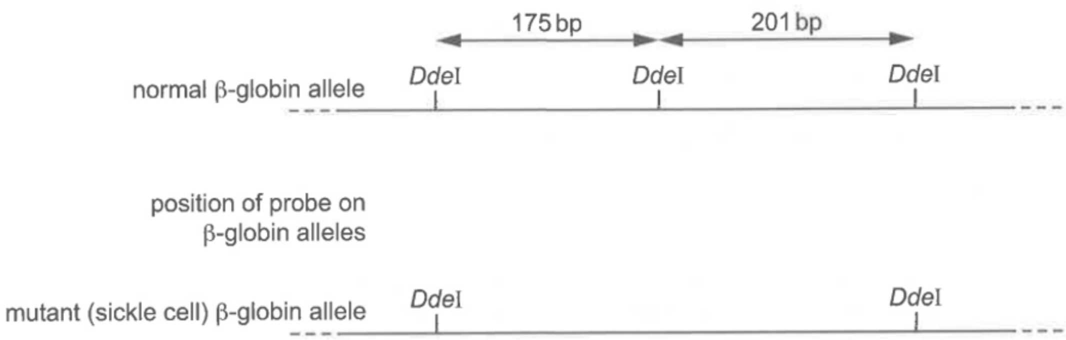

Fig. 6.3

(c) Using the information from Fig. 6.2,

(i) indicate in Fig. 6.3 the position of the probe that was used [1]

(ii) state which family members suffer from sickle cell anaemia.

.... [1]

[Total: 10]

Question 7 starts on page 18

7 Fig. 7.1 shows an animal cell in a stage of the mitotic cell cycle.

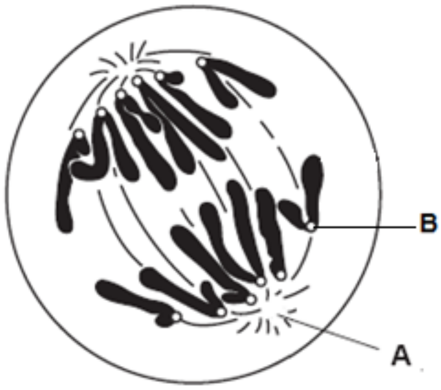

Fig. 7.1

(a) A pair of rod-like structures can be found in region **A**.

State **one other** feature of these structures and outline their role during mitosis in animal cells.

feature ....

....

role ....

....

[2]

(b) Identify structure **B** and outline its structure and function.

B ....

structure ....

....

function ....

....

[3]

(c) With reference to Fig. 7.1, state **two** observable differences between the behaviour of chromosomes in mitosis and meiosis.

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.... [2]

(d) The stages and checkpoints of the mitotic cell cycle are closely regulated. Explain the need to regulate the mitotic cell cycle tightly.

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.... [3]

[Total: 10]

8 The fruit fly, *Drosophila melanogaster*, feeds on sugars found in damaged fruits.

A fruit fly with normal features is described as wild type. It has red eyes and its wings are longer than its abdomen. There are mutant variations such as purple eyes or short (vestigial) wings.

Fig. 8.1 shows a wild type fruit fly and a mutant fruit fly with purple eyes and vestigial wings.

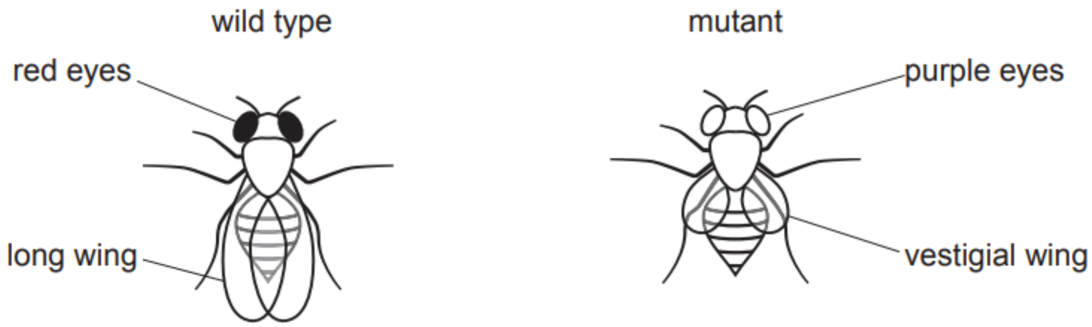

Fig. 8.1

- The genes coding for eye colour and wing length are located on the same chromosome.
- Allele **R** for red eyes is dominant to allele **r** for purple eyes.
- Allele **N** for long wings is dominant to allele **n** for vestigial wings.

(a) A wild type fruit fly, heterozygous for both genes, was crossed with a fruit fly that was homozygous recessive for both genes.

Table 8.1 is a summary of the cross.

Complete Table 8.1.

Table 8.1

	wild type parent		**double homozygous recessive parent**	
parental phenotype	....	x	....	
parental genotype	....	x	....	
offspring				
offspring genotype	....	....	....	....
offspring phenotype	....	....	....	....
number of offspring	1339	1195	151	154

[4]

(b) A statistical test showed that the results of the cross were significantly different from those expected.

State the name of the statistical test used and explain why it would be useful to carry out the test on these results. No calculations are required to answer this question.

statistical test ....

explanation ....

.... [2]

(c) Describe and explain the difference between the results of the cross in Table 8.1 and those expected.

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.... [4]

[Total: 10]

9 The red poppy, *Papaver rhoeas*, and several species of daisy often co-exist as weeds of wheat fields.

Fig. 9.1 shows changes in the percentage frequency of red poppies and daisies in an area of wheat fields over a six year period from 1998 to 2003. From 1985, the herbicide metsulfuron-methyl was used to control weeds in this area of wheat fields. This practice continued throughout the six year period.

1998 showed the first occurrence of a red poppy known as biotype X. This red poppy had a specific mutation not present in normal red poppies.

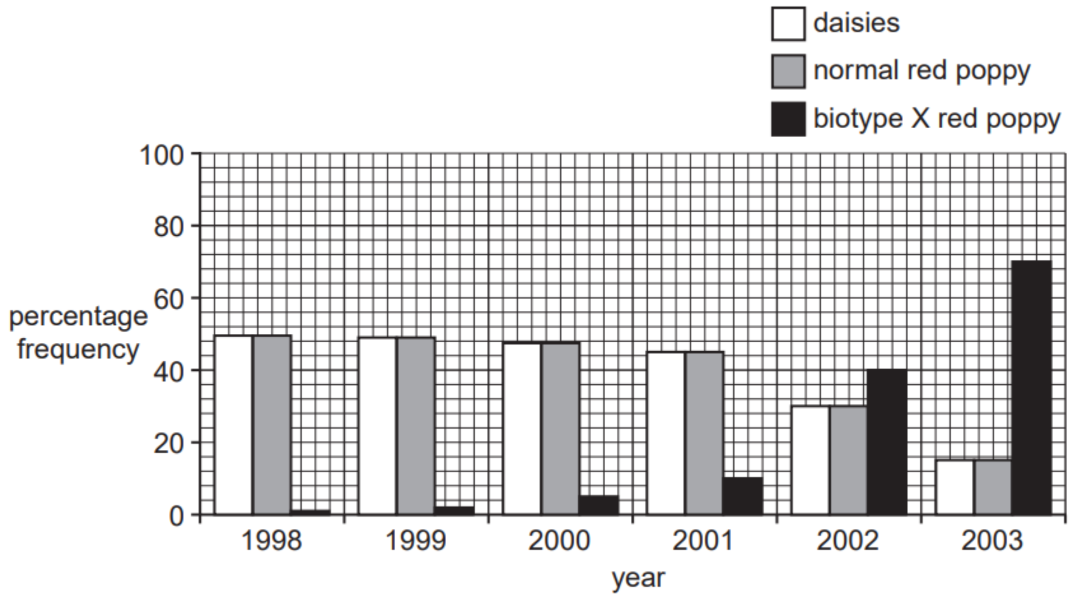

Fig. 9.1

(a) Describe how the percentage frequencies of red poppies and daisies changed over the six year period.

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.... [3]

(b) Suggest a reason for the distinct difference in percentage frequencies of normal red poppies and biotype X red poppy in the year 2003.

....

.... [1]

Although biotype X red poppy and normal red poppies continue to grow within the same geographical location, after numerous generations, biotype X red poppy may eventually become a different species from the normal red poppies.

(c) Suggest why biotype X red poppy may eventually become a different species from the normal red poppies.

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.... [2]

Daisies belong to the *Compositae* family, which is the largest family of flowering plants comprising more than 25 000 species. Many important evolutionary questions about the diversity in this family remain unanswered due to the lack of evidence to support major nodes of the phylogeny.

(d) Explain the advantages of using molecular methods to determine phylogenetic relationships within the *Compositae* family.

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.... [3]

Fossil evidence indicates that flowering plants first appeared about 125 million years ago and they have since been rapidly diversifying. The comparison of chloroplast genomes could provide insight into the evolution of flowering plants.

(e) Explain how evidence based on homologies identified in chloroplast genomes support Darwin's theory of evolution.

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.... [2]

Although plant fossils are usually rare compared to fossils of bones, teeth and shells, there are many great fossil sites in the world that have excellent preservation of plant materials.

(f) Suggest why plant fossils are usually rare.

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.... [1]

[Total: 12]

10 Measles is a common viral infection. A vaccine has been available for measles since the 1960s. There are vaccination programmes for many diseases including measles. Babies are born with a passive immunity to measles so the vaccine is not given in the first few months after birth.

(a) Explain how active immunity differs from passive immunity.

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.... [3]

The World Health Organisation (WHO) publishes data on the vaccination programmes for infectious diseases. The WHO recommends vaccination rates of over 90% of children.

Each health authority in a country reports its success in vaccinating children in their district. The WHO uses these figures to estimate the percentage of districts in each country that vaccinate 90% of children against measles.

The WHO also collects statistics on death rates of children under the age of 5 from all causes, including infectious diseases.

Fig. 10.1 shows these statistics for 24 countries for the year 2007.

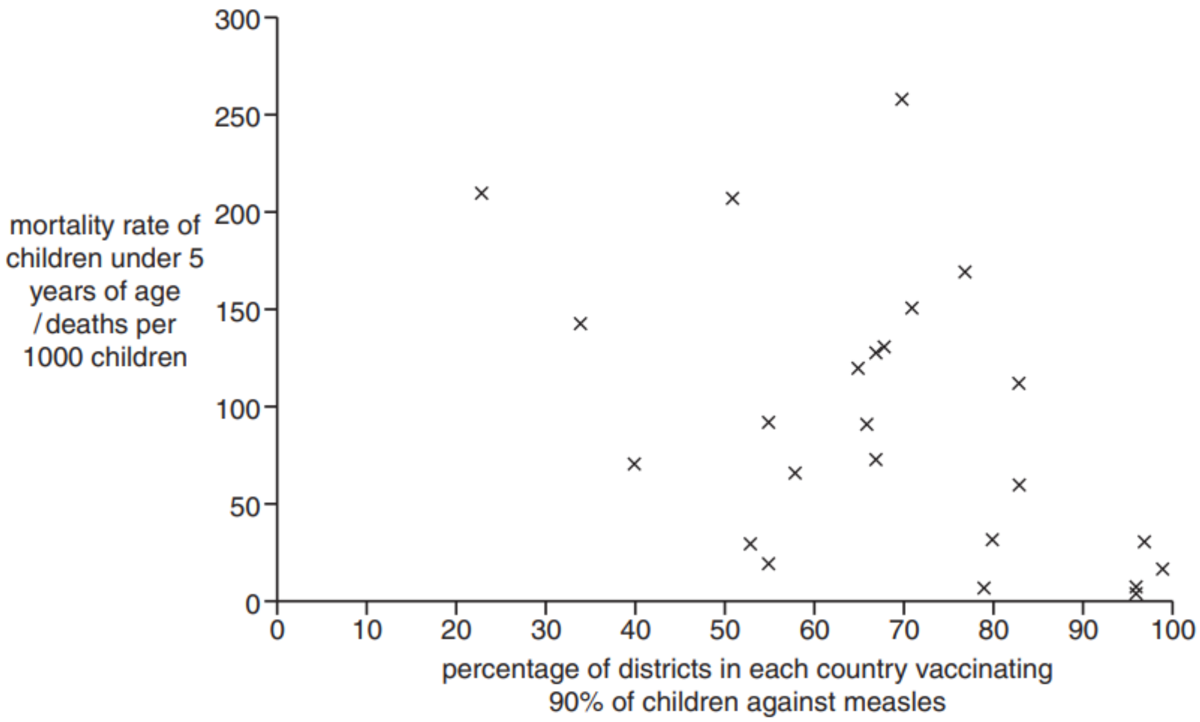

Fig. 10.1

(b) Use the information in Fig. 10.1 to explain why the WHO recommends immunisation of 90% of children.

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.... [2]

[Total: 5]

Question 11 starts on page 29

11 The concept of climate change and global warming has been of concern to scientists for many years.

(a) One way to collect data about atmospheric concentrations of greenhouse gases in the past is to study samples of ice from ice sheets in Antarctica. Ice samples from deep in the ice sheets were formed hundreds of thousands of years ago, while those near the surface were formed recently.

As ice forms, small bubbles of air are trapped in the ice. These air bubbles can be analysed to determine the concentration of carbon dioxide present. It is also possible to use chemical techniques to determine when the air bubbles were trapped.

Scientists studying climate change measured carbon dioxide concentrations in air bubbles from ice samples of known age, collected from near the surface.

Fig. 11.1 shows the concentration of carbon dioxide measured in air bubbles that were trapped in ice from 1959 to 1990. Direct measurements of atmospheric carbon dioxide from 1948 to 1978 are also shown.

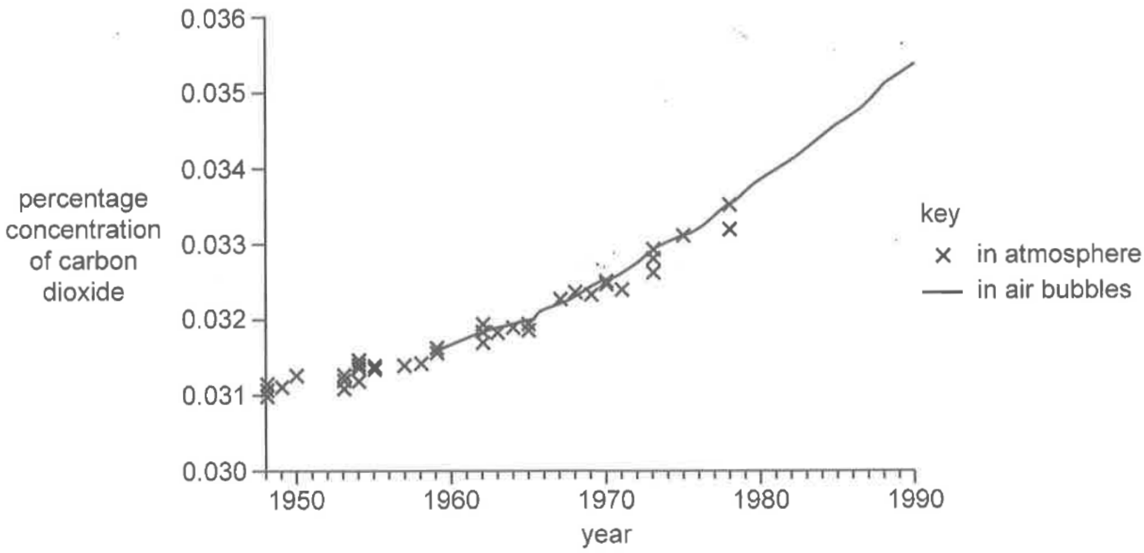

Fig. 11.1

With reference to the information provided, suggest and explain how climate change scientists can estimate atmospheric carbon dioxide concentrations 10 000 years ago.

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.... [2]

(b) Volcanic eruptions, which eject large volumes of gases such as water vapour and carbon dioxide to heights of 16 to 32 kilometres above the Earth's surface, have had an effect on the climate of the Earth.

Toba is a volcano in Sumatra that erupted approximately 12 000 years ago. Fig. 11.2 shows the location of Sumatra.

Fig. 11.2

Fig. 11.3 shows the changes to sea levels, compared to the present day, in the oceans around Greenland.

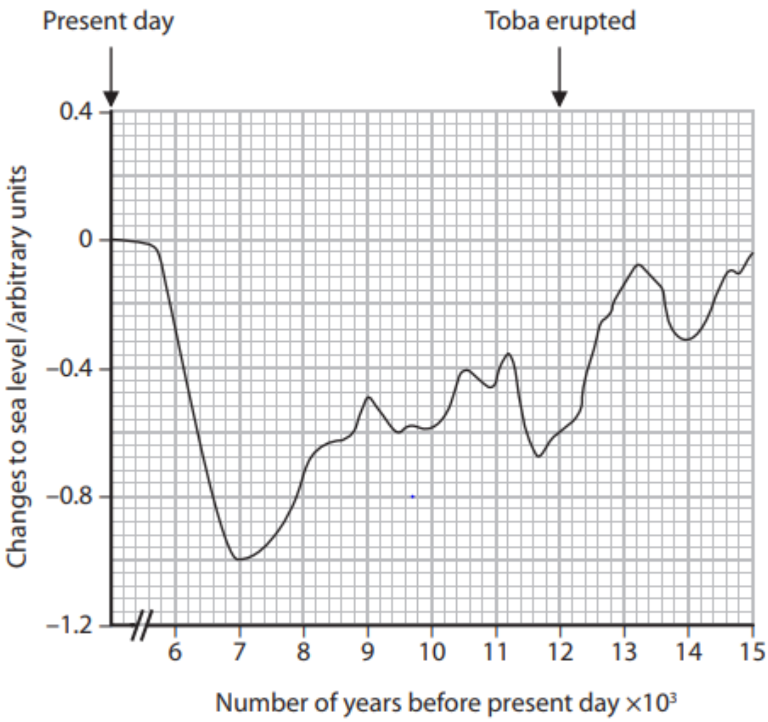

Fig. 11.3

It is claimed that the volcanic eruption of Toba caused a change in world climate.

(i) Describe the evidence in Fig. 11.3 that supports this claim.

....

.... [1]

(ii) Suggest why this claim may not be true.

....

....

....

.... [2]

[Total: 5]

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